

AI DEVELOPMENT WORKFLOW ASSIGNMENT

Part 1: Short Answer Questions

Problem Definition The AI problem chosen is Predicting Student Dropout Rates. The objective is to identify students at risk of dropping out before it happens. The objectives include improving student retention, providing early interventions, and optimizing resource allocation. Stakeholders are university administrators and academic advisors. The key performance indicator (KPI) is the model's accuracy in predicting dropout likelihood.

Data Collection & Preprocessing Two data sources would be the student information system (SIS) and the learning management system (LMS). A potential bias in the data could arise from unequal data representation across socioeconomic backgrounds. The preprocessing steps include handling missing data, encoding categorical variables, and normalizing numerical features.

Model Development A Random Forest model is selected because it handles both categorical and numerical data efficiently and is robust against overfitting. Data will be split into 70% training, 15% validation, and 15% testing sets. Two hyperparameters to tune are the number of trees and maximum tree depth to balance bias and variance.

Evaluation & Deployment Evaluation metrics include accuracy and F1-score. Accuracy measures overall performance, while F1-score ensures balance between precision and recall. Concept drift occurs when relationships in data change over time; it would be monitored through continuous model retraining and performance tracking. A technical challenge during deployment would be scalability for large datasets.

Part 2: Case Study Application

Problem Scope The hospital aims to develop an AI system that predicts patient readmission risk within 30 days of discharge. Objectives are to reduce readmission rates, improve patient outcomes, and optimize hospital resources. Stakeholders include doctors, hospital administrators, and patients.

Data Strategy Data sources include electronic health records (EHRs), demographic data, and discharge summaries. Ethical concerns involve patient privacy and algorithmic bias. The preprocessing pipeline will include handling missing values, encoding categorical variables such as diagnosis, and feature engineering by combining lab results into risk indicators.

Model Development A Logistic Regression model is chosen for interpretability and reliability in medical predictions. A confusion matrix is constructed: True Positives = 70, False Positives = 20, False Negatives = 30, True Negatives = 80. Precision = $70 / (70 + 20) = 0.78$, Recall = $70 / (70 + 30) = 0.70$.

Deployment Integration will occur via an API into the hospital's EHR system, allowing real-time predictions during patient discharge. Compliance with healthcare regulations such as HIPAA will be ensured through encryption, anonymization, and secure data storage.

Optimization To address overfitting, cross-validation and L2 regularization will be used.

Part 3: Critical Thinking

Ethics & Bias Biased training data may lead to unequal treatment recommendations or inaccurate risk scores for minority groups. One mitigation strategy is to audit data regularly for bias and apply fairness-aware machine learning techniques.

Trade-offs There is a trade-off between interpretability and accuracy. While deep learning models may achieve higher accuracy, simpler models like Logistic Regression are preferred in healthcare.

due to transparency. Limited computational resources may restrict use of large neural networks, favoring lightweight, interpretable models.

Part 4: Reflection & Workflow Diagram

Reflection The most challenging part was defining the problem and ensuring data quality. With more time, I would explore more robust feature engineering and additional data sources to improve prediction accuracy.

Workflow Diagram:

